

INNOVATIVE APPLICATIONS OF INDUSTRIAL ROBOTICS

Robotic Cell for Automated Helmet Visor Testing and Visual Inspection

Authors

Cosimo De Luca · Matteo Casazza · Francesco Giglio · Marco Giavazzi

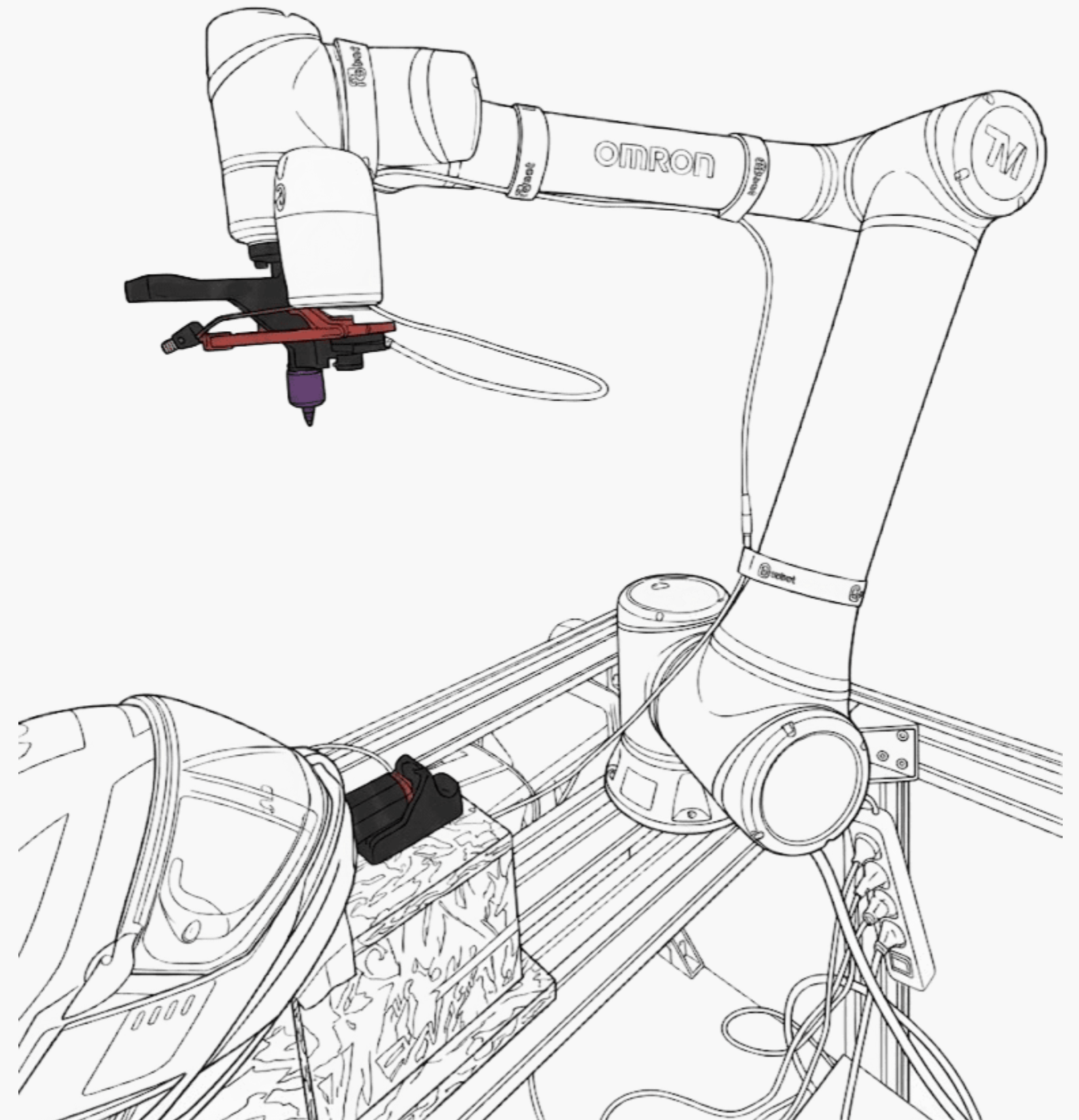
Alberto Feltrin · Donato Miraglia

Supervisor

Giovanni Bianchi

Academic Year

2025 / 2026



POLITECNICO
MILANO 1863



THE GOAL

PROJECT OBJECTIVES

Six pillars — each one anchors a section of this deck

01

CONSTRAINING

Constrain the helmet
rigidly & repeatably

02

END EFFECTOR

Integrated tool:
handling, vision &
marking

03

HANDLING

Robot control & Visor
tasks

04

ACQUISITION

Capture the surface
with the stereo-camera

05

DETECTION

Locate surface
defects automatically

06

INSPECTION & MARKING

Identify, refine and mark
each detected defect

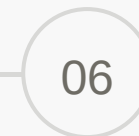
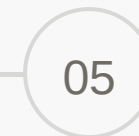
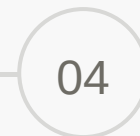
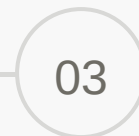




SECTION 01 / 06

Constraining

Constrain the helmet rigidly and repeatably



CONSTRAINING

END EFFECTOR

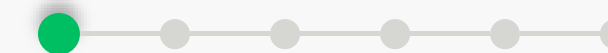
HANDLING

ACQUISITION

DETECTION

INSPECTION & MARKING





THE OBJECT

ANATOMY OF THE HELMET

Key features the cell must handle



01 Visor tab

Tab that help for the opening of the visor

02 Sun-visor levers

Slider that opens the inner sun visor and button to close it

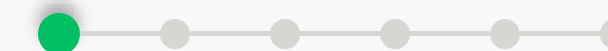
03 Chin bar release button

A mechanism to release the lower part of the helmet

04 Inspection zone

Zone of the helmet we have to inspect to find eventual damage

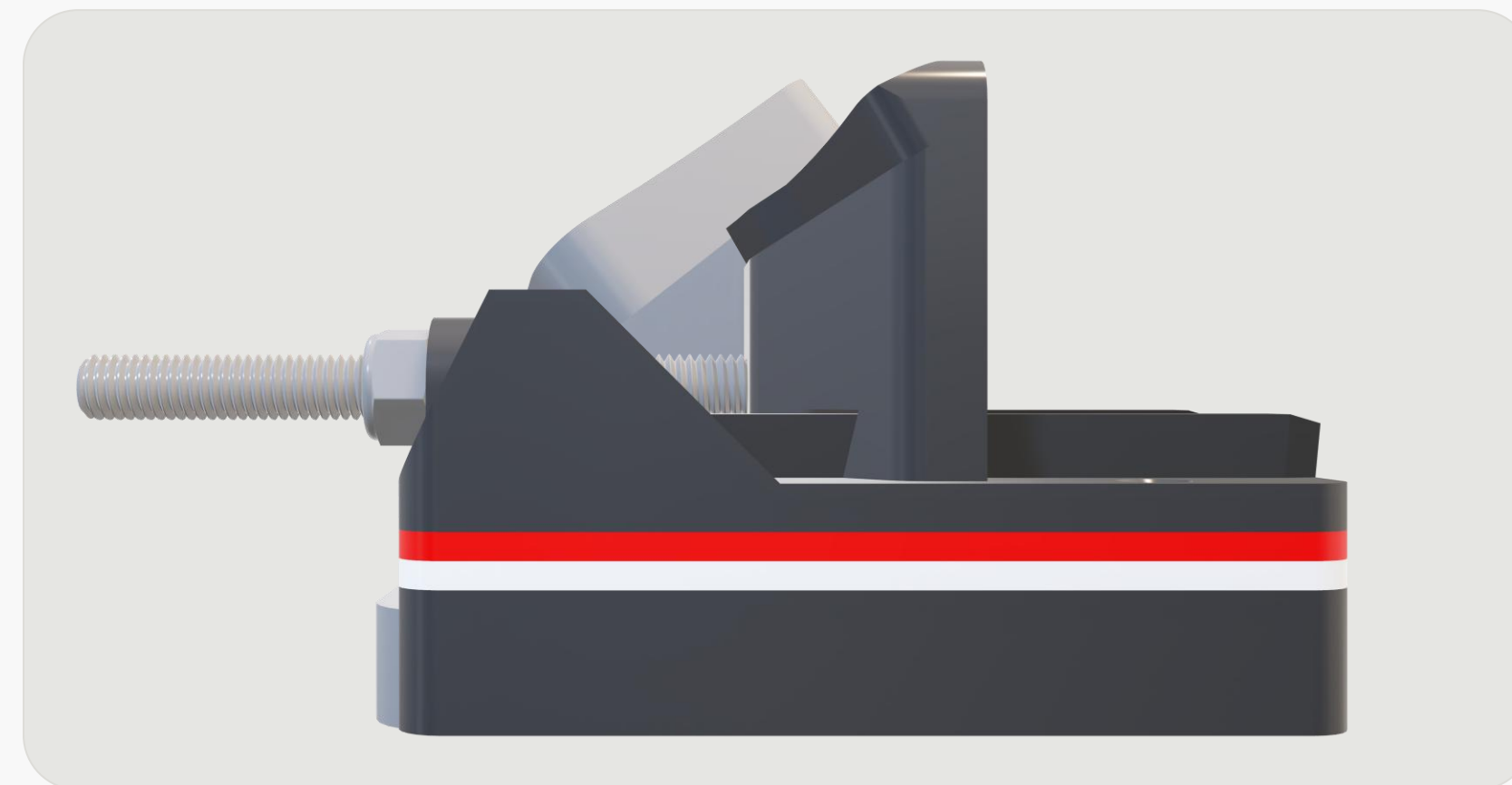
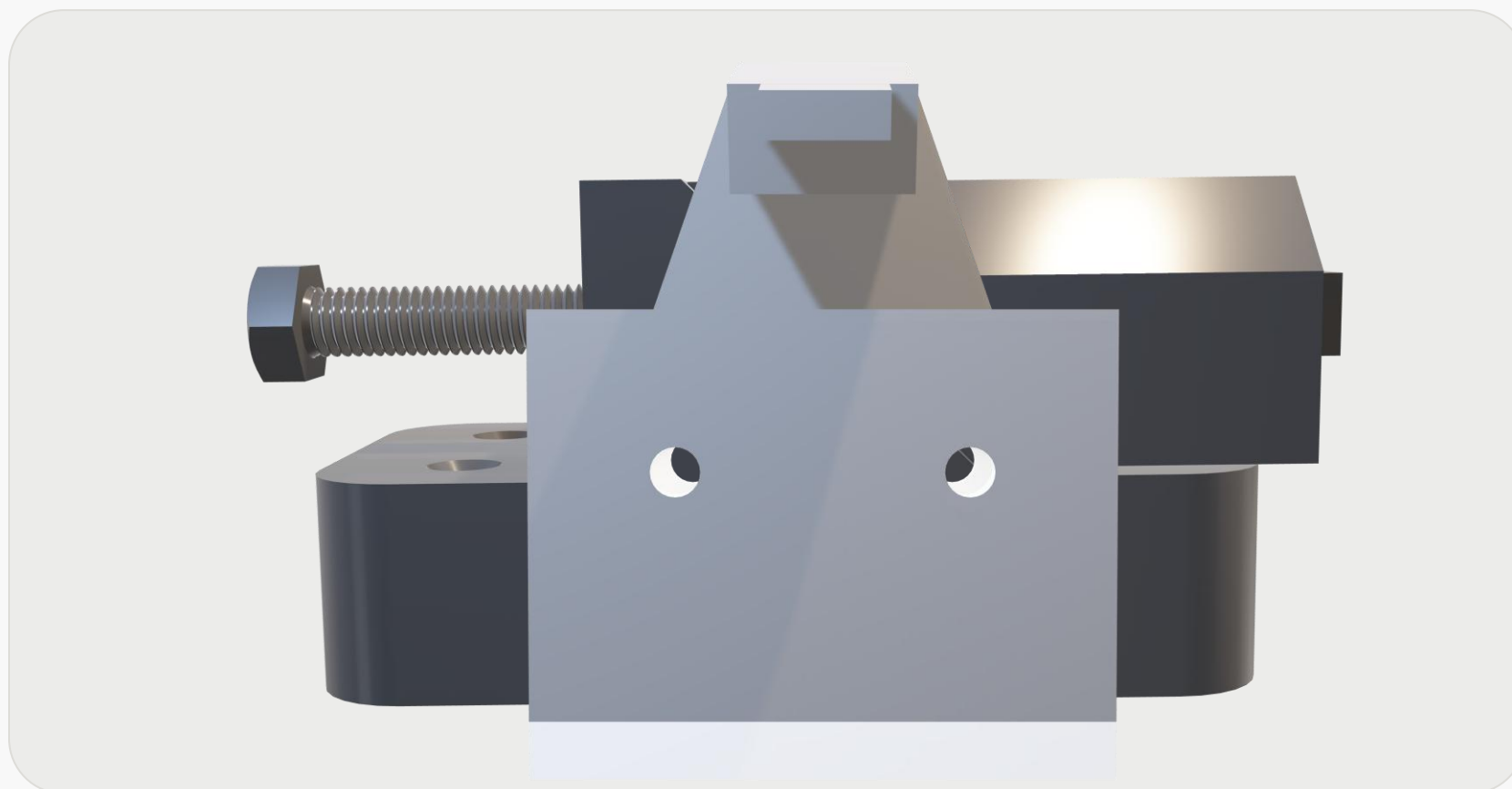




THE DESIGN

DESIGNING THE FIXTURE

Modular, tool-free and repeatable by design



01

STABILITY

Rigid under external loads

02

REPEATABILITY

Same position every cycle

03

ACCESSIBILITY

Clear approach for visors,
marking & to complete the
tasks

04

MODULAR

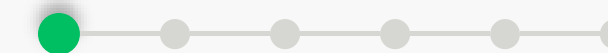
Adjustable - fits all helmet
sizes

05

FAST-MOUNT

Mounts & dismounts rapidly
and easily

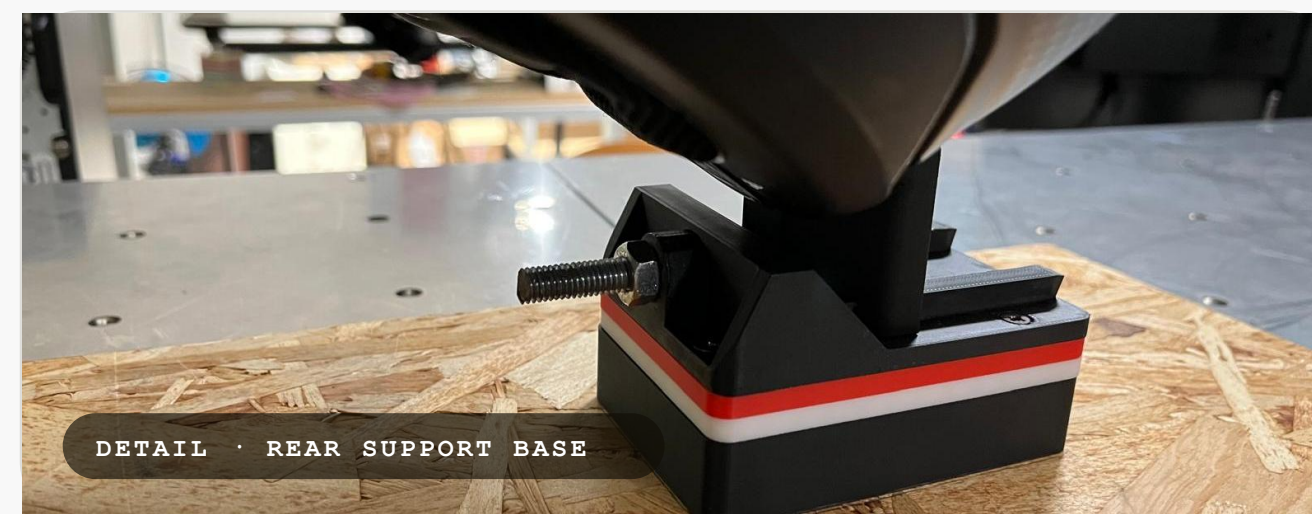




IMPLEMENTATION

THE CLAMPING SYSTEM

Realised fixture — front push-clamp and rear support base



- *Key idea: the helmet remain in the exact same position on every cycle.*

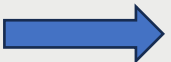




FORCES



Reaction Forces: 

Applied Forces: 

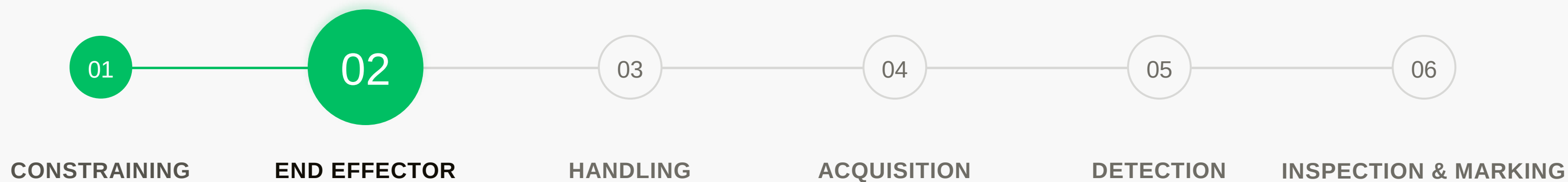


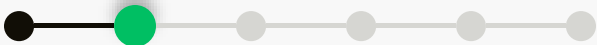


SECTION 02 / 06

End Effector

One integrated tool for handling, vision and marking

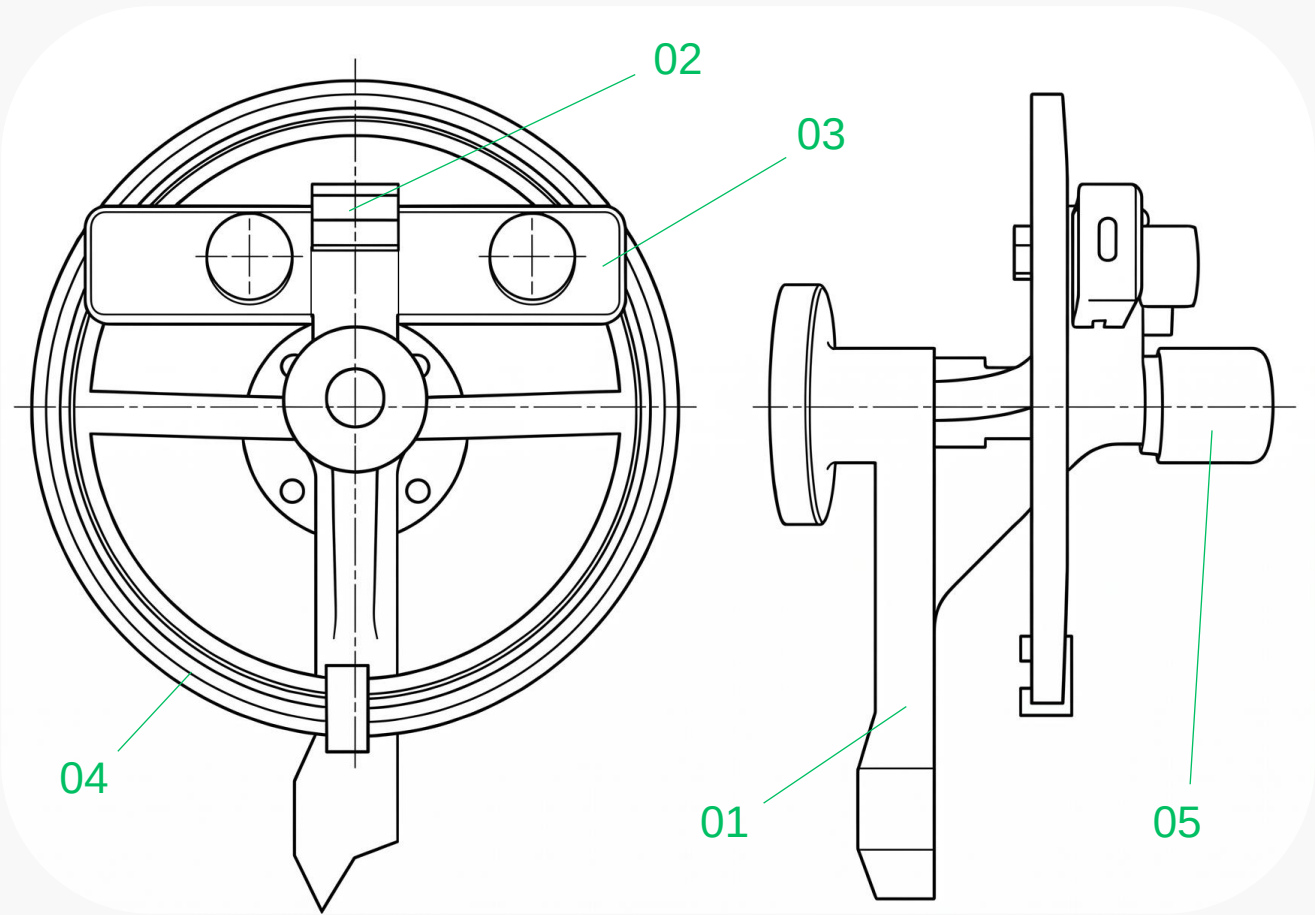




THE DESIGN

END-EFFECTOR DESIGN

CAD model and bill of materials



BILL OF MATERIALS

#	COMPONENT	QTY
01	Base + Integrated Finger Rigid frame, robot mount & finger — One printed piece	1
02	Stereo-camera support Holds the ZED mini	1
03	ZED mini stereo-camera 3D surface acquisition	1
04	LED ring + support Illumination system	1
05	Marker system Cap, Marker, Spring – Marking defects	1

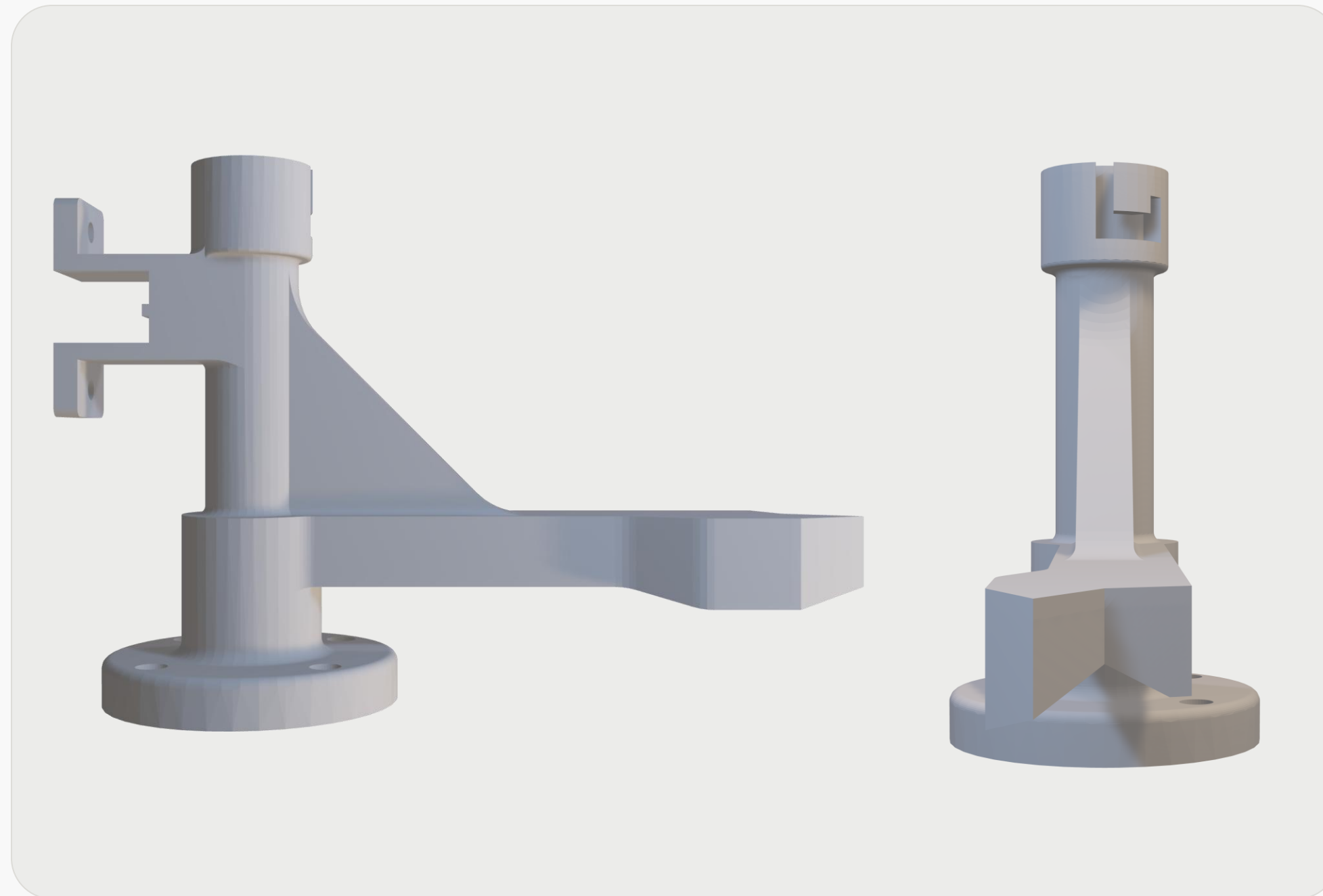




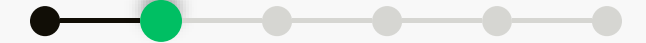
IMPLEMENTATION

ASSEMBLING THE END-EFFECTOR

Progressive build-up — one component at a time



End Effector · Stage 2 / 6



STEP 1 / 5 · COMPONENT ADDED

BASE + INTEGRATED FINGER

FUNCTION

Structural backbone of the entire end-effector, perform manual tasks and house camera and marker

RATIONALE

3D-printed design with optimized layer orientation

ADVANTAGE

Maximum strength — no joint or fastener to fail

● Base + integrated finger

- ☐ Stereo-camera support
- ☐ LED support
- ☐ Marker system

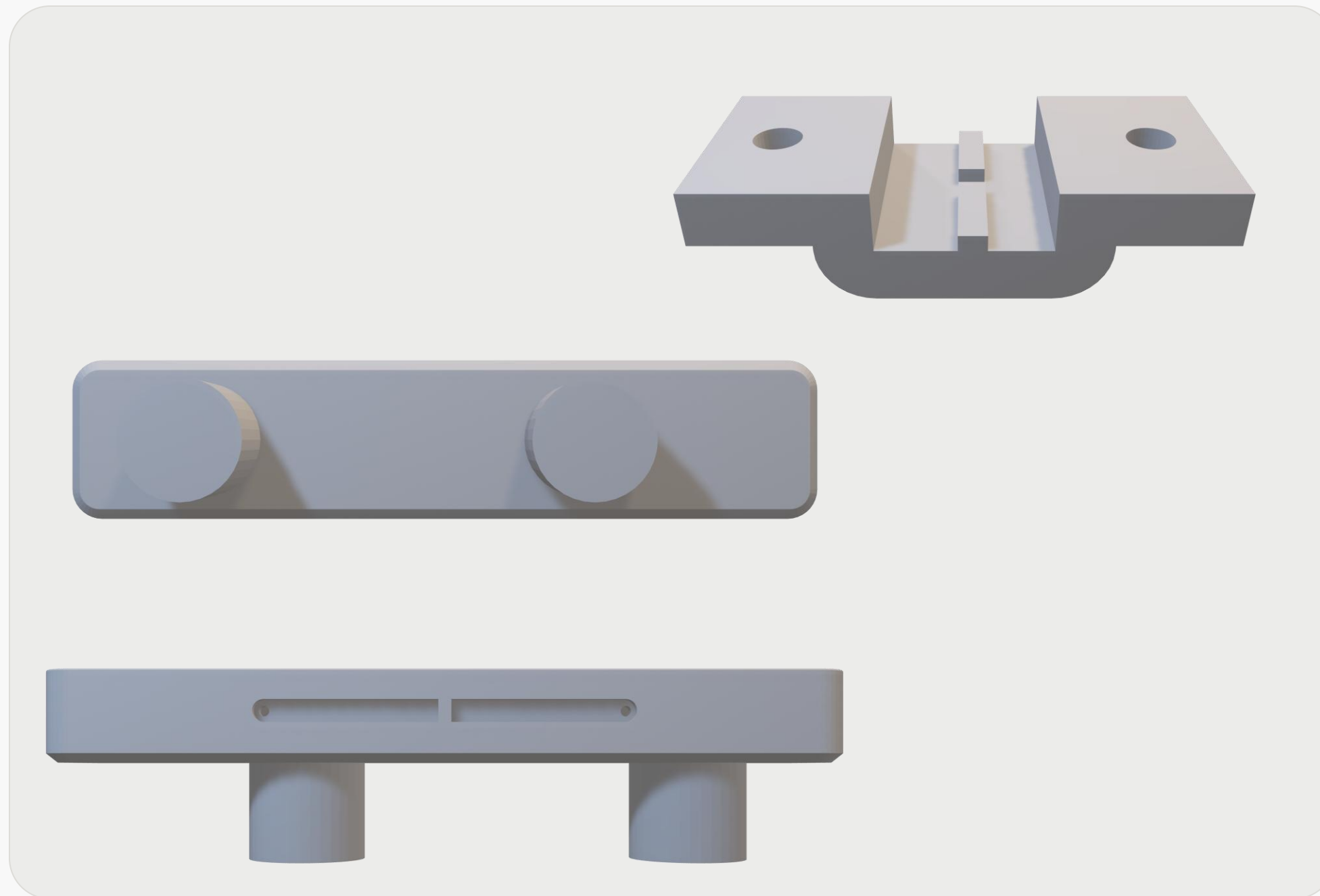




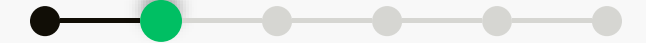
IMPLEMENTATION

ASSEMBLING THE END-EFFECTOR

Progressive build-up — one component at a time



End Effector · Stage 2 / 6



STEP 2 / 5 · COMPONENT ADDED

STEREO-CAMERA SUPPORT

FUNCTION

Holds the ZED mini that captures the 3D surface

RATIONALE

Custom matching grooves designed to interlock perfectly with the camera's original housing

ADVANTAGE

Self-align assembly, easy to disassemble

● Base + integrated finger

● Stereo-camera support

○ LED support

○ Marker system

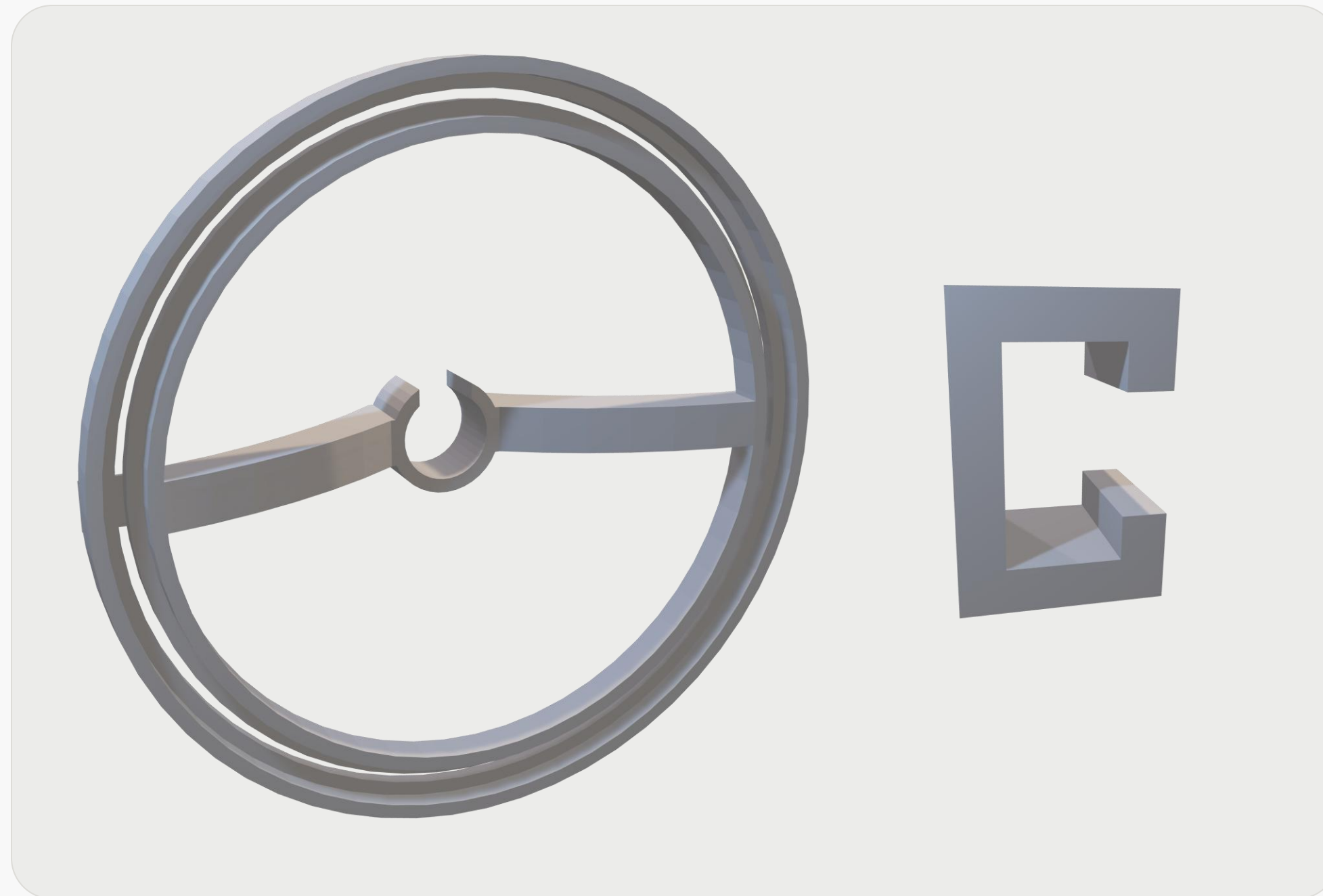




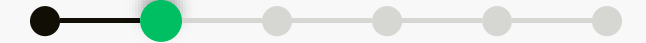
IMPLEMENTATION

ASSEMBLING THE END-EFFECTOR

Progressive build-up — one component at a time



End Effector · Stage 2 / 6



STEP 3 / 5 · COMPONENT ADDED

LED SUPPORT

FUNCTION

Mounts the LED ring for uniform illumination

RATIONALE

Snap-fit mechanical locking

ADVANTAGE

Easy to disassemble, no impact on the robot's workspace

- Base + integrated finger
- Stereo-camera support
- LED support
- Marker system

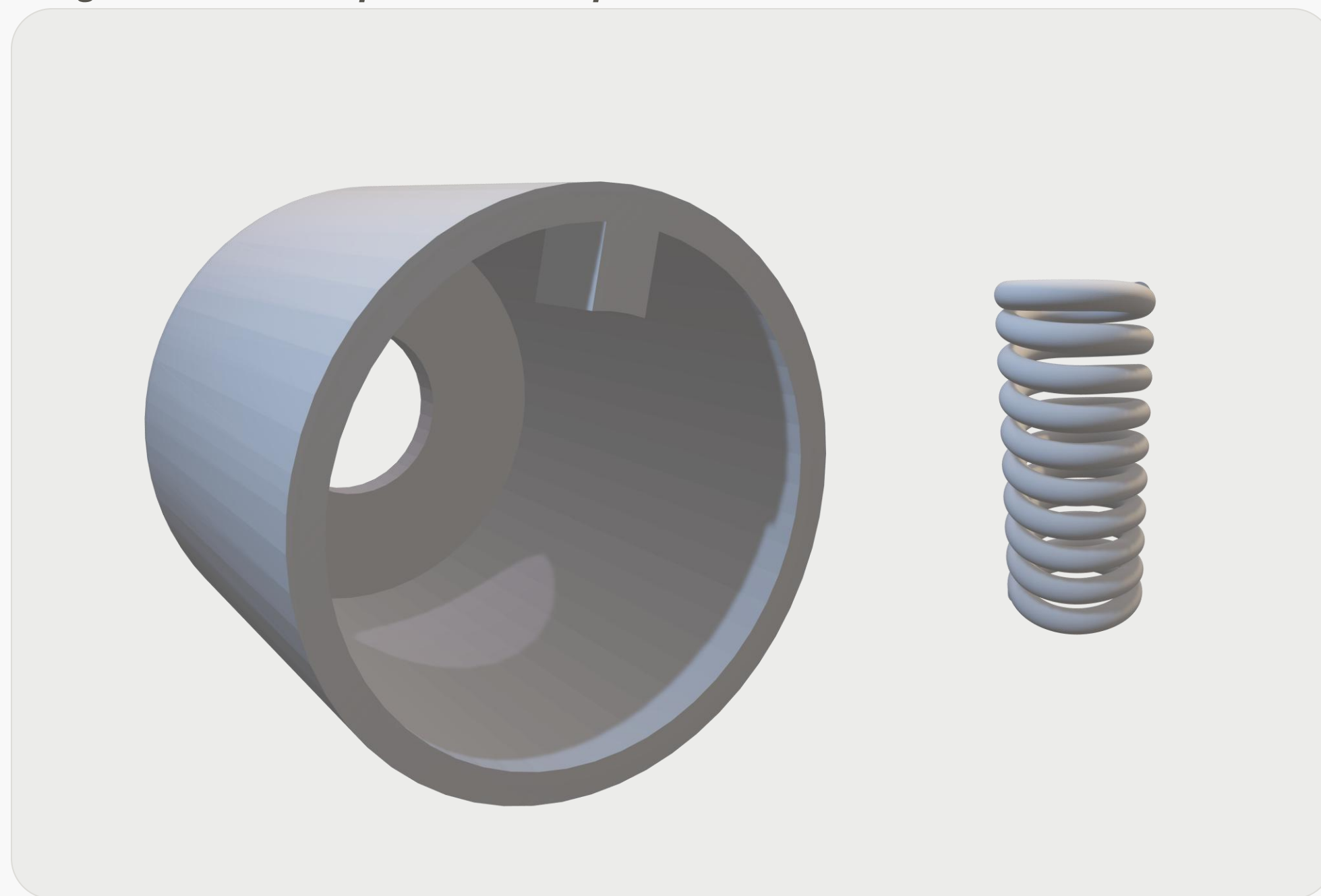




IMPLEMENTATION

ASSEMBLING THE END-EFFECTOR

Progressive build-up — one component at a time



End Effector · Stage 2 / 6



STEP 4 / 5 · COMPONENT ADDED

MARKER SYSTEM

FUNCTION

Secures the marking pen and dampens contact forces

RATIONALE

Axial spring-loaded compliance, locking system with main body

ADVANTAGE

Retractable pen, Easy to mount and dismount

- Base + integrated finger
- Stereo-camera support
- LED support
- Marker system

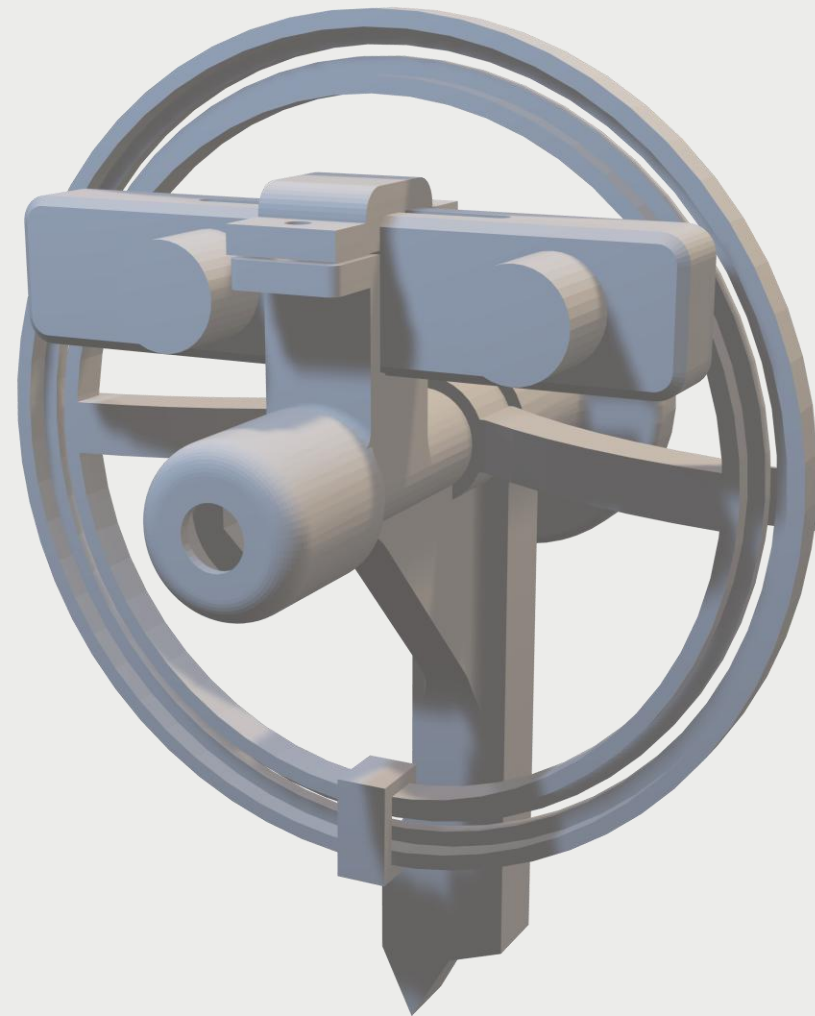




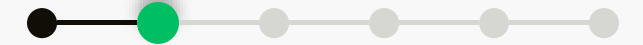
IMPLEMENTATION

ASSEMBLING THE END-EFFECTOR

Progressive build-up — one component at a time



End Effector · Stage 2 / 6



STEP 5 / 5 · COMPLETE

COMPLETE ASSEMBLY

FUNCTION

Handling, vision, lighting & marking in one tool

RATIONALE

One end-effector for the entire cycle

ADVANTAGE

No tool changes — faster inspection

- Base + integrated finger
- Stereo-camera support
- LED support
- Marker system

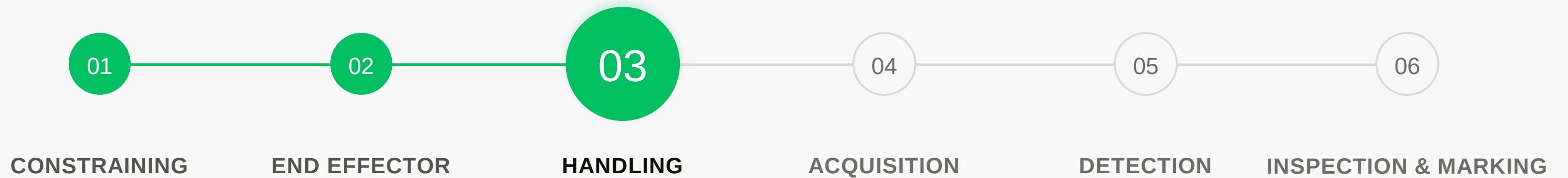




SECTION 03 / 06

Handling

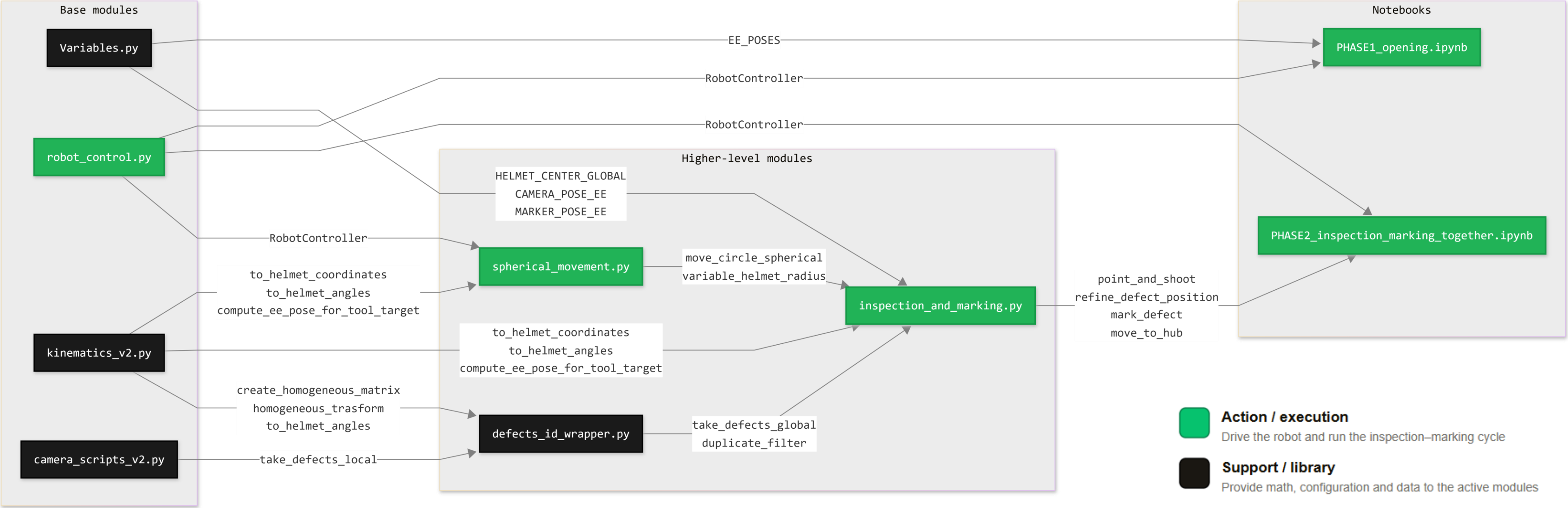
Robot control & Visor tasks



HANDLING

SOFTWARE ARCHITECTURE

Modular architecture, orchestrated by two workflows





HANDLING

ROBOT CONTROL

Listen Node commands, Modbus TCP feedback



01 COMMAND INTERFACE

A single class sends motion scripts to the Techman Listen Node and reads the pose back over Modbus TCP.

02 SYNCHRONOUS & BLOCKING

The program waits until the robot reaches the target within a tolerance, by continuously polling the pose

03 MOTION PRIMITIVES

Joint moves, Cartesian point-to-point, linear moves on the TCP, and 3D arcs

04 SAFETY

Home pose, emergency stop and a movement timeout





HANDLING

REFERENCE FRAMES & THE MATRIX H

One 4×4 carries rotation and translation

$$\mathbf{H} = \begin{bmatrix} \begin{matrix} \mathbf{R} \\ \mathbf{T} \end{matrix} \\ \begin{matrix} 0 & 0 & 0 & 1 \end{matrix} \end{bmatrix}$$

The matrix $\mathbf{H}$ is a 4x4 matrix. The top-left 3x3 block is labeled $\mathbf{R}$ and contains elements r_{11}, r_{12}, r_{13} in the first row, r_{21}, r_{22}, r_{23} in the second row, and r_{31}, r_{32}, r_{33} in the third row. The top-right 3x1 block is labeled $\mathbf{T}$ and contains elements t_x, t_y, t_z . The bottom row consists of four elements: 0, 0, 0, and 1. A dashed horizontal line separates the top three rows from the bottom row, and a dashed vertical line separates the 3x3 rotation block from the 3x1 translation block.

$$P_{(global)} = H \cdot P_{(local)}$$

01 FOUR REFERENCE FRAMES

Robot base - End-Effector (TCP) - tool (camera and marker, each with its own fixed offset) - helmet

02 ONE MATRIX H (4×4)

Rotation $\mathbf{R}$ (3×3) + translation $\mathbf{T}$ (3×1) in a single block takes a point from local to global

03 DIRECT & INVERSE

The direct chain $H_{(cam \rightarrow global)} = H_{(ee \rightarrow global)} \cdot H_{(cam \rightarrow ee)}$ brings a camera-space defect into the robot base frame.

The inverse H^{-1} does the opposite, global $\rightarrow$ local, we use it to strip the tool offset and recover the EE pose: $H_{ee} = H_{(target)} \cdot H_{(tool \rightarrow ee)}^{-1}$





HANDLING

ROTATION MATRIX ↔ ANGLES

Z-Y-X (Cardan) convention, aligned to the Techman controller

CONSTRUCTION

$$R = R_z(r_z) \cdot R_y(r_y) \cdot R_x(r_x)$$

Z-Y-X intrinsic $\equiv$ X-Y-Z extrinsic — matches the robot controller's Euler convention

EXTRACTION

$$s_y = \sqrt{(r_{11}^2 + r_{21}^2)}$$

$$r_x = \text{atan2}(r_{32}, r_{33})$$

$$r_y = \text{atan2}(-r_{31}, s_y)$$

$$r_z = \text{atan2}(r_{21}, r_{11})$$

$$R = \begin{bmatrix} \cos(r_z)\cos(r_y) & \cos(r_z)\sin(r_y)\sin(r_x) - \sin(r_z)\cos(r_x) & \cos(r_z)\sin(r_y)\cos(r_x) + \sin(r_z)\sin(r_x) \\ \sin(r_z)\cos(r_y) & \sin(r_z)\sin(r_y)\sin(r_x) + \cos(r_z)\cos(r_x) & \sin(r_z)\sin(r_y)\cos(r_x) - \cos(r_z)\sin(r_x) \\ -\sin(r_y) & \cos(r_y)\sin(r_x) & \cos(r_y)\cos(r_x) \end{bmatrix}$$

01 ATAN2, ALL FOUR QUADRANTS

Resolves the angle with the correct sign and avoids division by zero

02 SINGULARITIES

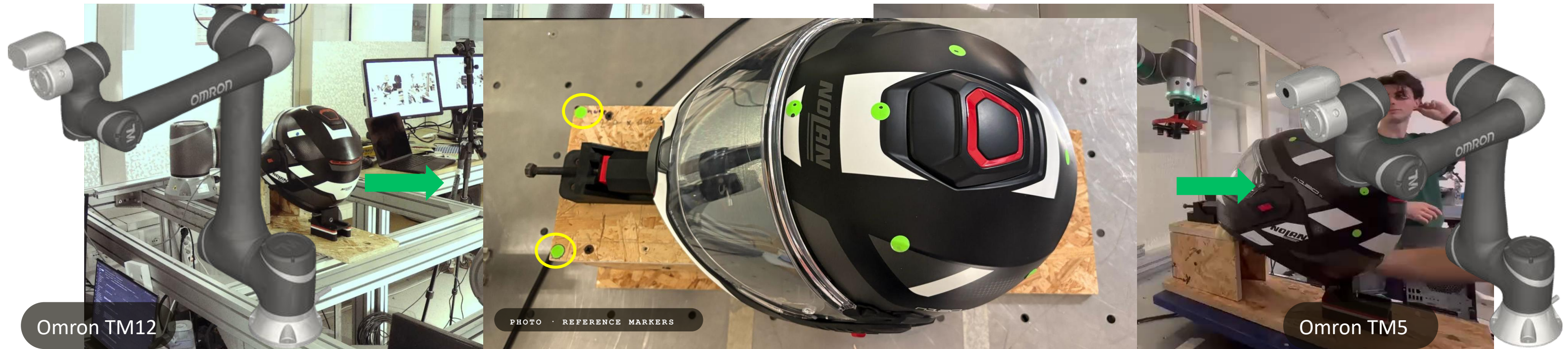
When $s_y \approx 0$ is pushed into zones the robot cannot reach



HANDLING

FROM OUR ROBOT TO ANY ROBOT

Generalising the poses so the cell is Adaptable



01

ABSOLUTE POSES

Hand-taught poses, valid only for that cell and that robot, not universal

02

MARKER-FRAME POSES

Each taught pose is re-expressed relative to a fixed reference frame on the base fixture (a marker) as:

$$\widetilde{P}_{loc} = H_{base}^{-1} \widetilde{P}_{taught}$$

03

RE-PROJECTION

Given the new marker pose, every pose is mapped back to the robot base as:

$$\widetilde{P}_{new} = H_{new} H_{base}^{-1} \widetilde{P}_{taught}$$

04

ADAPTABILITY

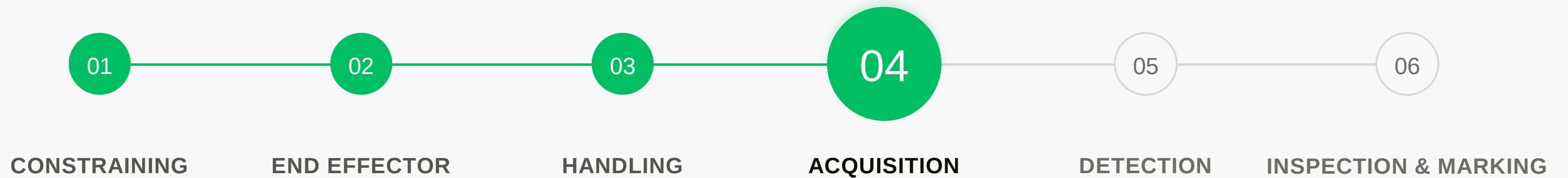
For a new cell or robot, just re-measure the marker pose. The whole opening/closing sequence re-projects automatically



SECTION 04 / 06

Acquisition

Capture the surface with the stereo-camera





ACQUISITION

AUTOMATED HELMET INSPECTION SYSTEM

3D Computer Vision & Robotic Kinematics for Defect Control

01 HELMET CENTER

The core coordinate

02 STEREO-CAMERA

Hardware configuration: Eye-in-Hand-system (camera rigidly attached to the EE)

03 3D SURFACE SCANNING

Continuos, automated 3D surface scanning ensuring perpendicularity





ACQUISITION

ORIENTATION CONSTRAINT: LOCAL Z-AXIS TRACKING

Ensuring Constant Optical Perpendicularity on the Helmet Center

EE POSE

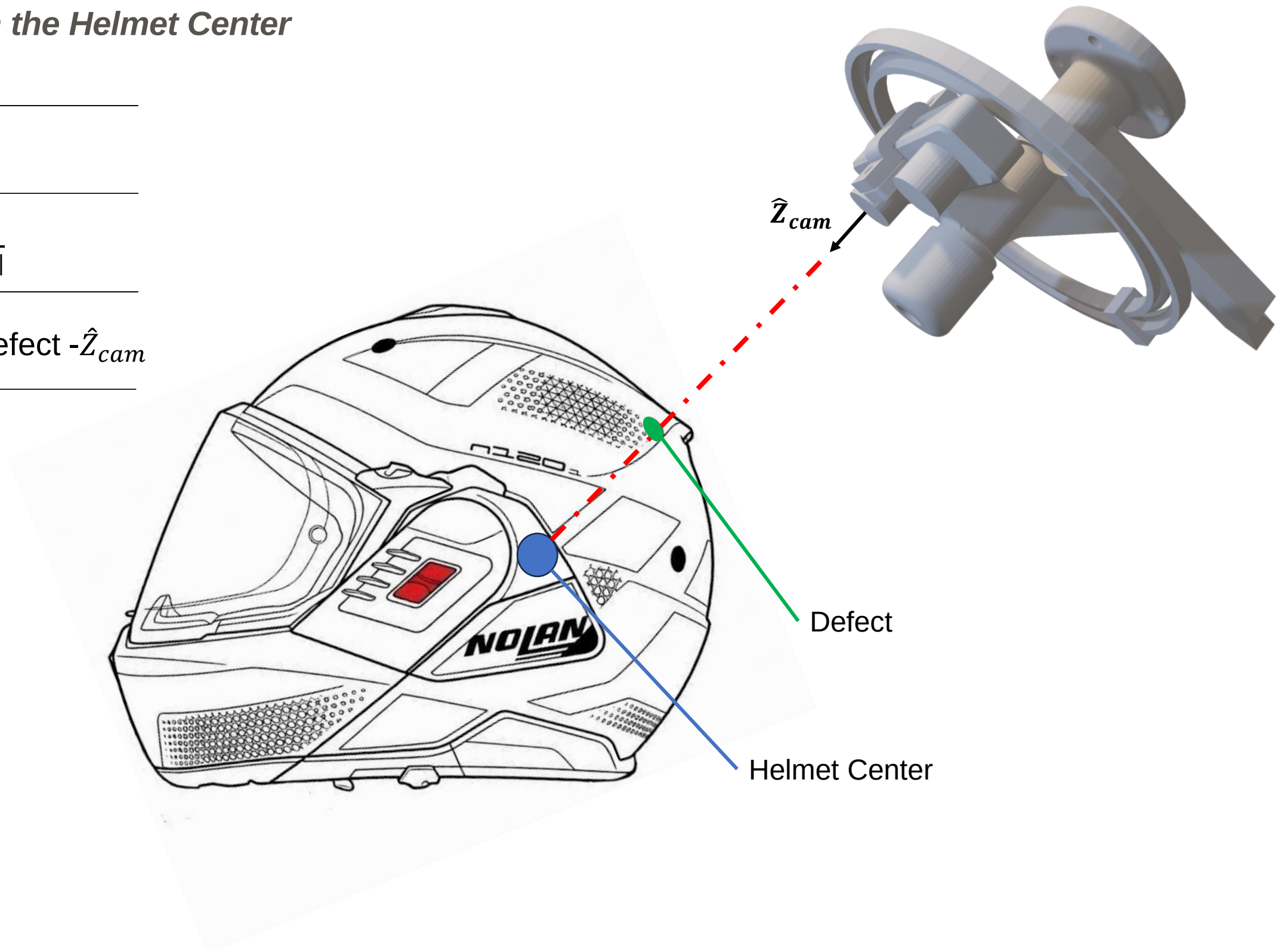
Defined by $p = [x, y, z]^T$

Z_{cam} ALIGNMENT

Camera Versor : $\hat{Z}_{cam} = \frac{-p}{\|p\|}$

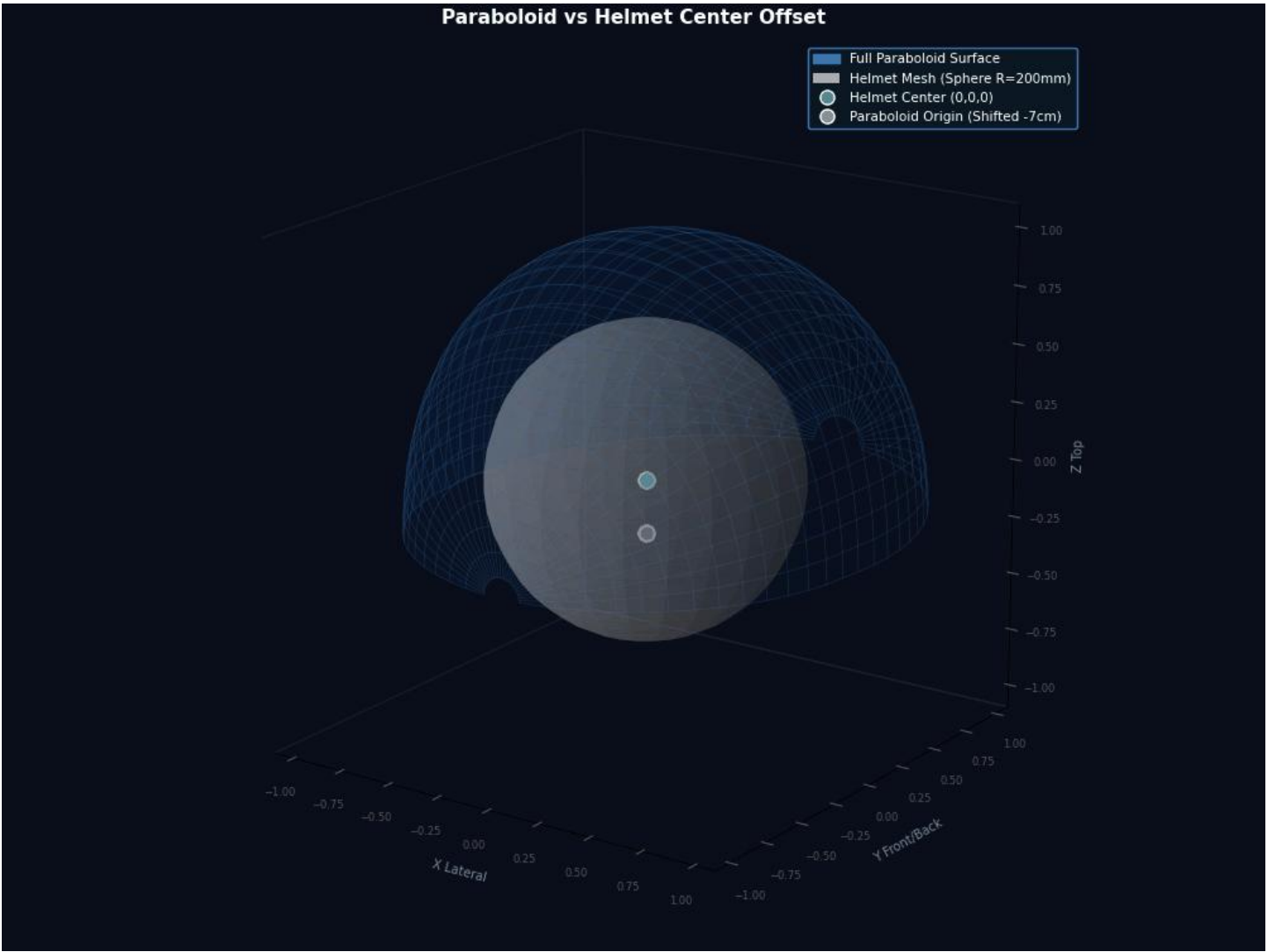
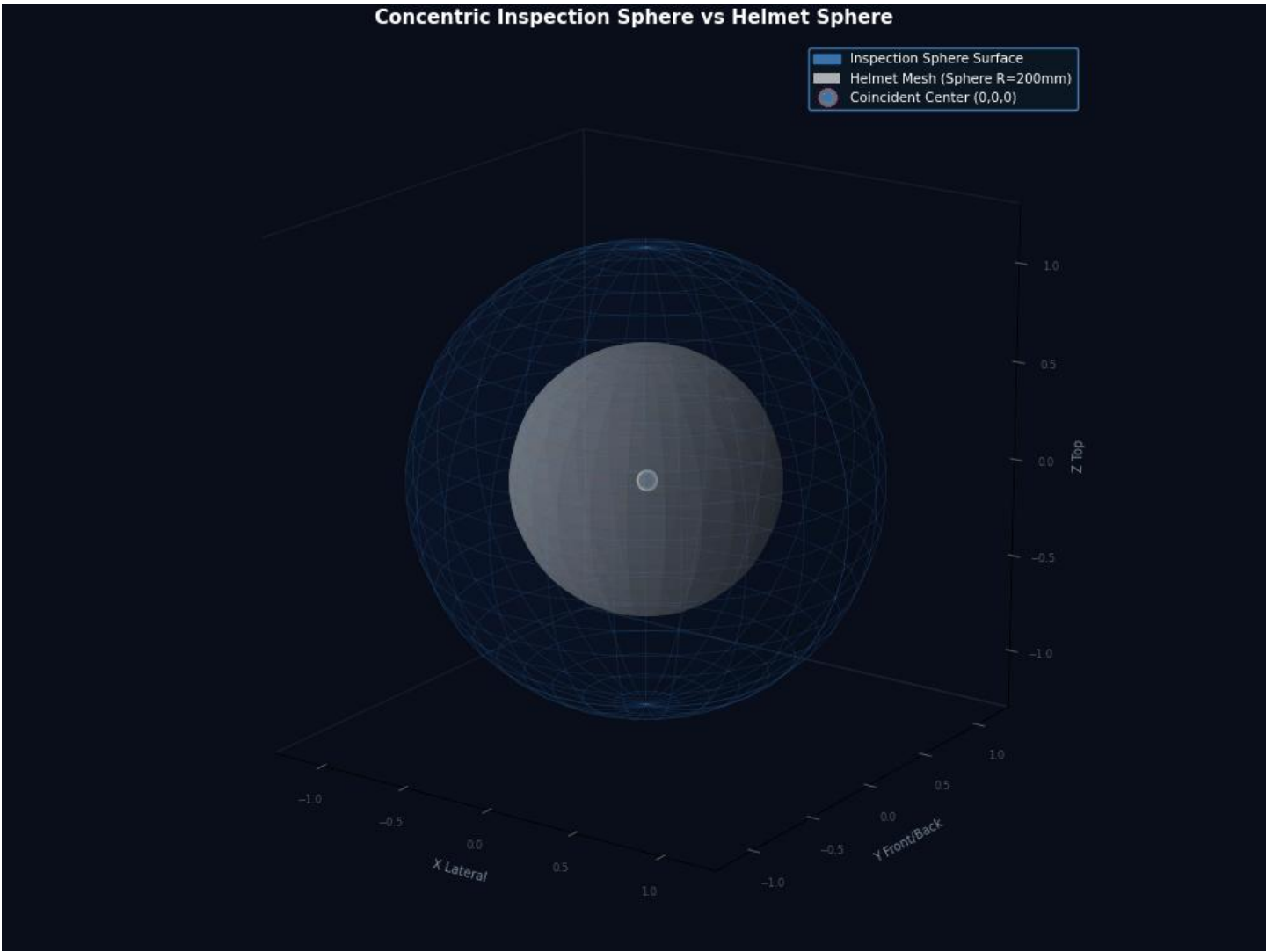
ADVANTAGE

Alignment Helmet Center – Defect $-\hat{Z}_{cam}$



ACQUISITION

TRAJECTORY GENERATION: VARIABLE HELMET RADIUS (VHR)



01

THE PROBLEM

Standard spheres trigger upper-shell collisions or miss lower helmet collar

02

SOLUTION

Paraboloid flaring shape that opens up as elevation decreases

03

DESIGN RADIUS

$$\begin{aligned} R_{APEX} &= 380\text{mm} \\ R_{SIDE/BACK} &= 300\text{mm} \\ R_{FRONT} &= 450\text{mm} \end{aligned}$$

04

TRIGONOMETRIC TRANSITION

Based on angle α (azimuth) and β (elevation)



ACQUISITION

COLLISION AVOIDANCE & VIRTUAL ENVIROMENT VALIDATION

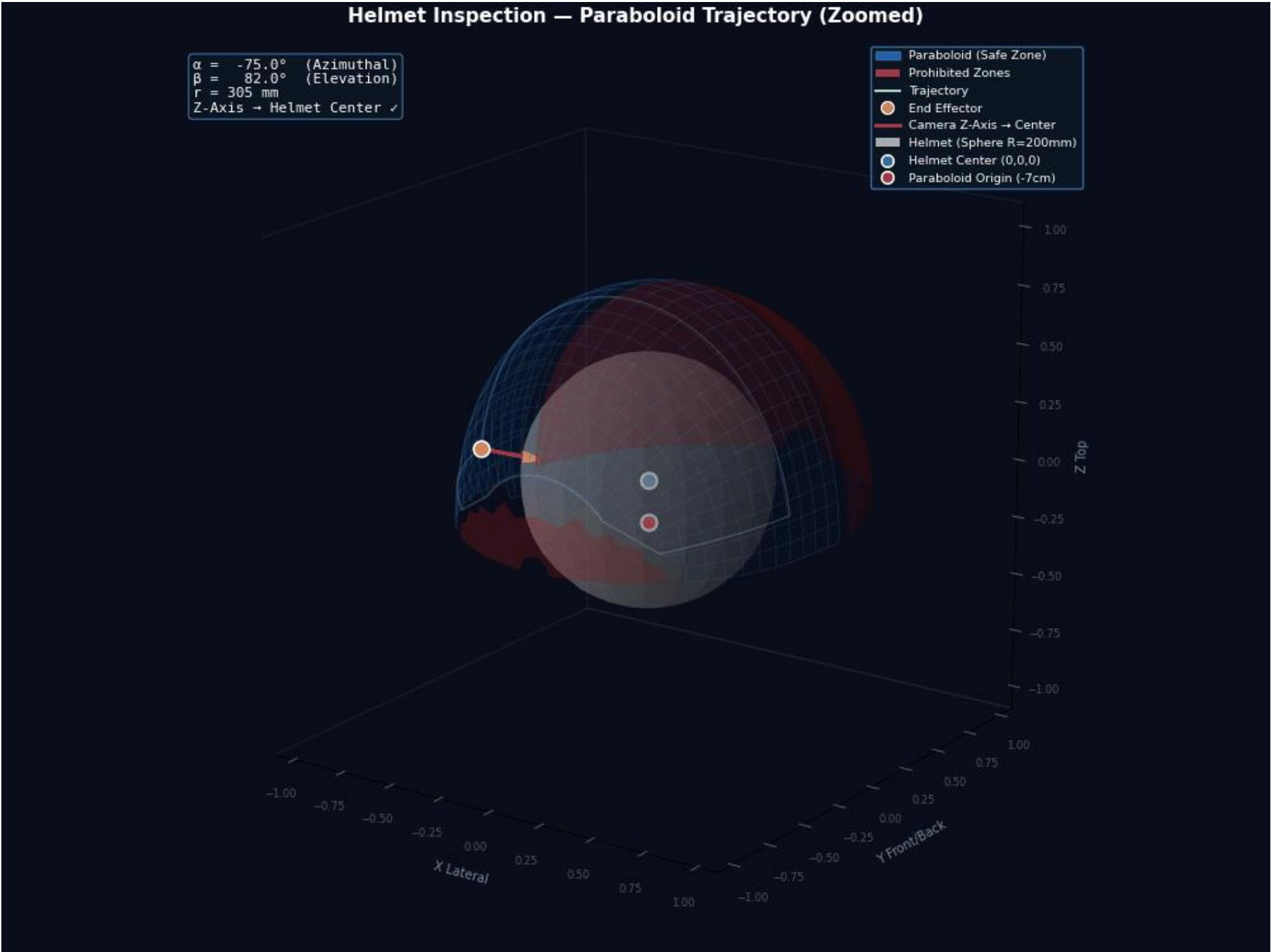
Bridging Algirithmic Safety with Digital Twin Validation

UNSAFE ZONES

01	Gimbal Lock Prevention	$ \alpha > 89^\circ$
02	Low Boundary Collision	$\beta < 20 + 25 \left(\frac{\alpha}{90}\right)^2$
03	Out-of-Bounds Limit	$\beta > 110^\circ$

DIGITAL TWIN PARAMETERS

01	Helmet modeled as a sphere	$R = 200mm$
02	Paraboloid centre	$[0, 0, -70]$
03	Scaling Normalization	$1/R_{APEX}$





ACQUISITION

IMAGE ACQUISITION

Stereo-camera capture images during the scan



RGB gives the 2D evidence; XYZ gives the depth information.



01

RGB IMAGES

Used for colour masks and 2D defect extraction

02

XYZ POINT CLOUD

Used to estimate the defect position in camera frame

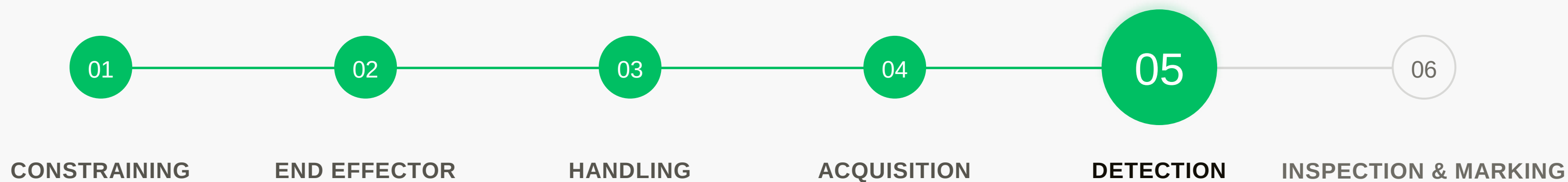




SECTION 05 / 06

Detection

Locate the surface defects automatically





DETECTION

2D DEFECT DETECTION: FROM MASK TO CENTROID

HSV segmentation is cleaned and converted into isolated defect candidates

01 MASK GENERATION

Direct green mask

Green known colour

BGR → HSV → green threshold

Inverse anomaly mask

For generic unknown colour

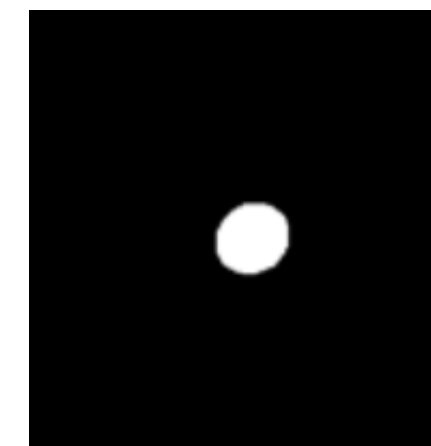
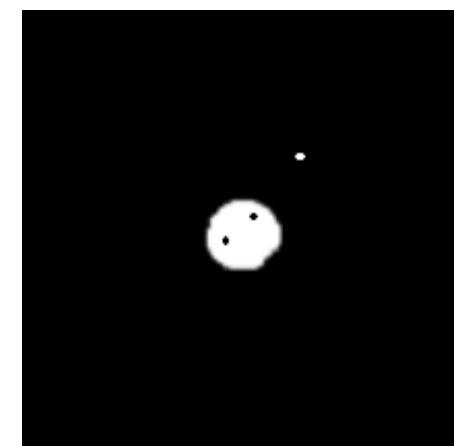
BGR → HSV → helmet colours threshold → inversion of the mask

02 MASK CLEANING

Attention radius → useful field of view

Opening → removes isolated noises

Closing → fills small holes





DETECTION

2D DEFECT DETECTION: FROM MASK TO CENTROID

HSV segmentation is cleaned and converted into isolated defect candidates

03 BLOB FILTERING

Find external contours
Reject small blobs by area threshold

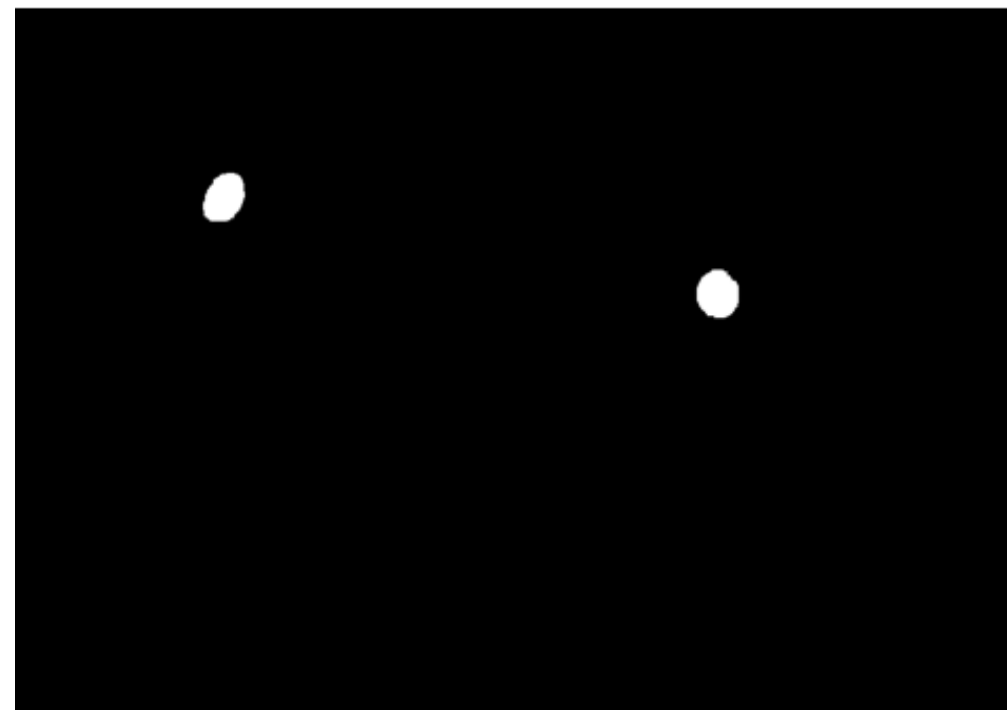
04 CENTROID & ISOLATED MASK

Compute area and centroid from image
Store one isolated mask for defect

Scatto 1 - RGB



Scatto 1 - Mask



Scatto 1 - RGB





DETECTION

FROM 2D MASK TO 3D DEFECT POSITION

White mask pixels are projected into the ZED point cloud and averaged

01 POINT CLOUD LOOKUP

From white pixels of isolated defect mask

Pixel $(x,y) \rightarrow [X_c, Y_c, Z_c]$ in camera frame

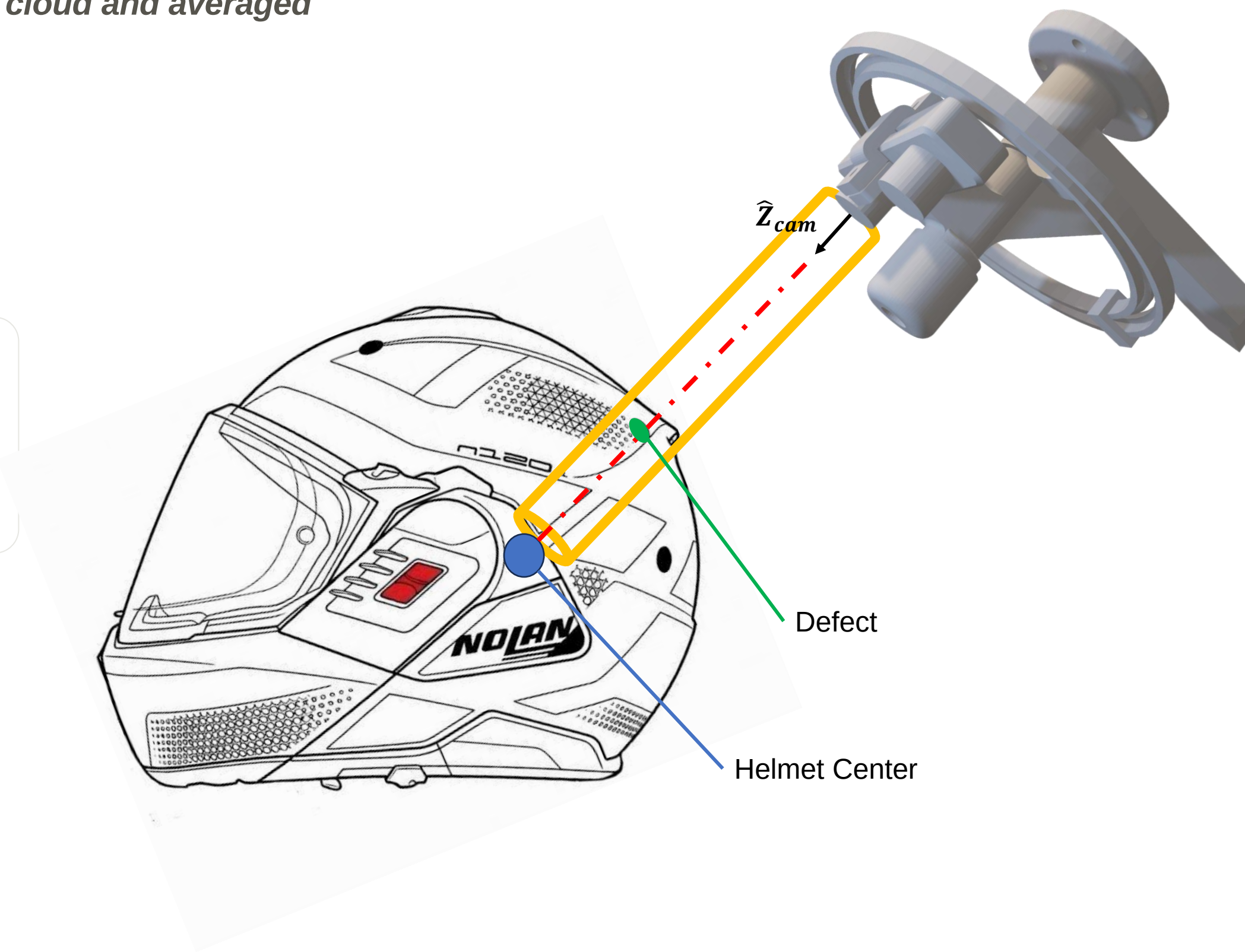
02 VALID XYZ SAMPLES AND CYLINDER FILTER

Reject failed, NaN or invalid depth values

Apply filter (lateral radius and depth range)

03 MEAN 3D POINT

Mean of 3D points of one defect in camera frame





DETECTION

FROM CAMERA FRAME TO HELMET TARGET

The 3D defect points are transformed from camera frame to the robot global frame and then expressed relative to the helmet

01 CAMERA-TO-GLOBAL TRANSFORM

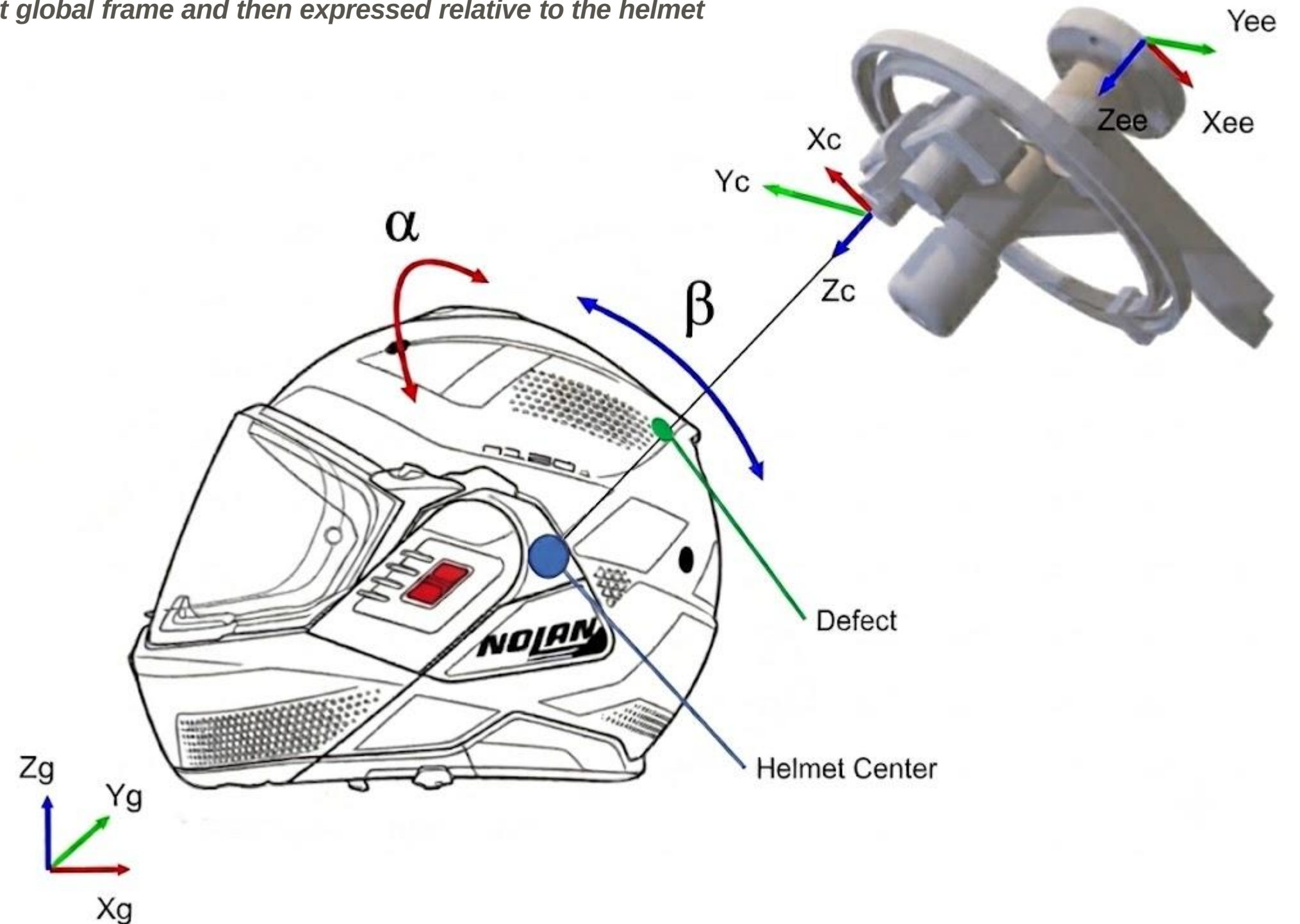
$$H_{(cam \rightarrow global)} = H_{(ee \rightarrow global)} \cdot H_{(cam \rightarrow ee)}$$

02 GLOBAL DEFECT POSITION

$$P_{(global)} = H_{(cam \rightarrow global)} \cdot P_{(cam)}$$

03 HELMET-RELATIVE TARGET

$$[r, \alpha, \beta] = \text{to_helmet_angles}(\text{pos3d_global}, \text{helmet_center})$$

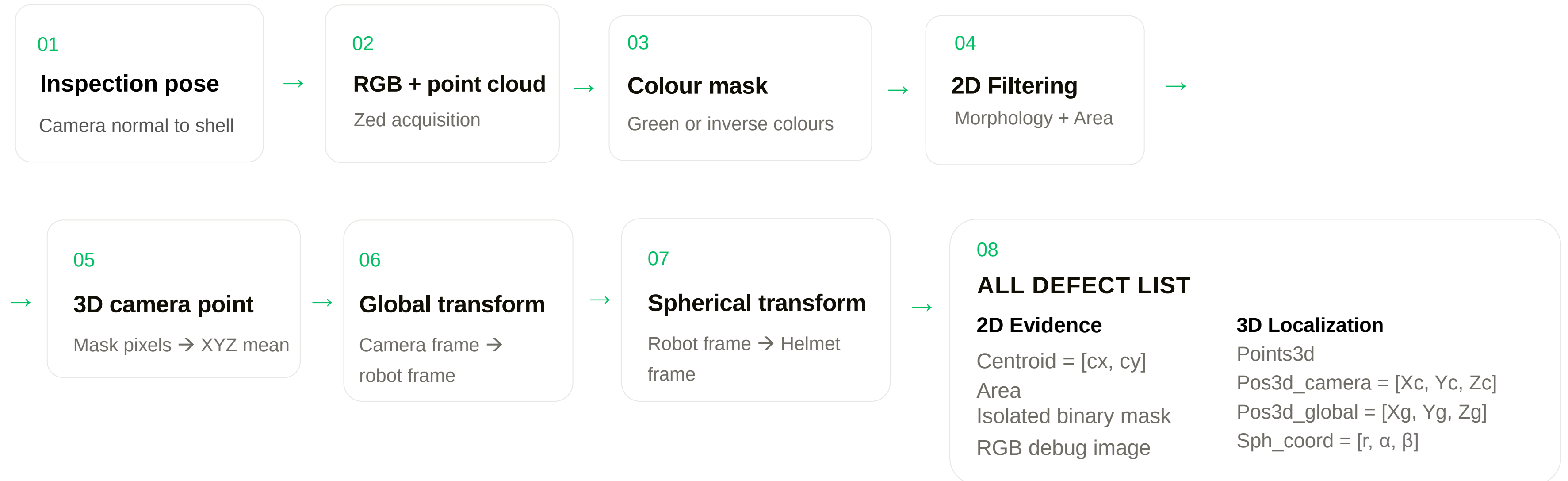




DETECTION

COMPLETE VISION PIPELINE

RGB image and point cloud are converted into helmet-relative defect coordinates

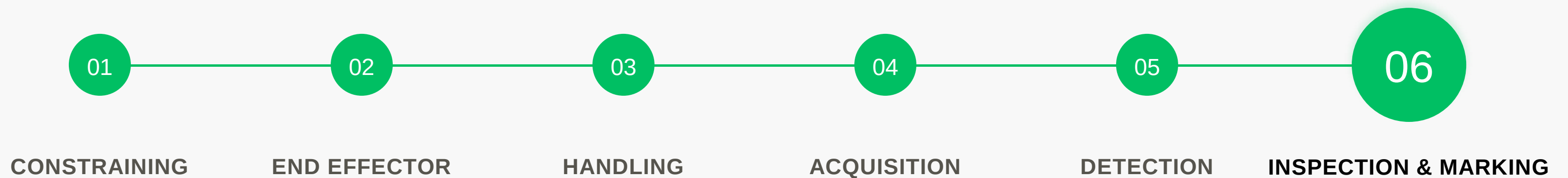




SECTION 06 / 06

Inspection & Marking

Identifying, refining and marking each detected defect on the shell





INSPECTION & MARKING

TOOL POSITIONING

Controlling the tool position via the robot pose

TOOL POSITIONING

We need to position the tools
attached to the end effector

CAMERA_POSE_EE,
MARKER_POSE_EE

ROBOT CONTROL

All the base movement commands
accept the robot's end effector pose

EE_POSE = [x,y,z,rx,ry,rz]

`kin.compute_ee_pose_for_tool_target(p_obj, r_obj, tool_pose_ee):`

$$H_{\text{tool_obj}} = H_{\text{ee_obj}} @ H_{\text{tool_ee}}$$

$$H_{\text{ee_obj}} = H_{\text{tool_obj}} @ (H_{\text{tool_ee}})^{-1}$$

$$H_{\text{ee_obj}} = \begin{bmatrix} R_{\text{ee}} & | & t_{\text{ee}} \\ \hline 0 & 0 & 0 & | & 1 \end{bmatrix}$$





INSPECTION & MARKING

HELMET INSPECTION

Initial inspection of the helmet to find all defects

INSPECTION POSITIONS

We manually set multiple inspection positions that will cover all of the helmet's inspectionable surface

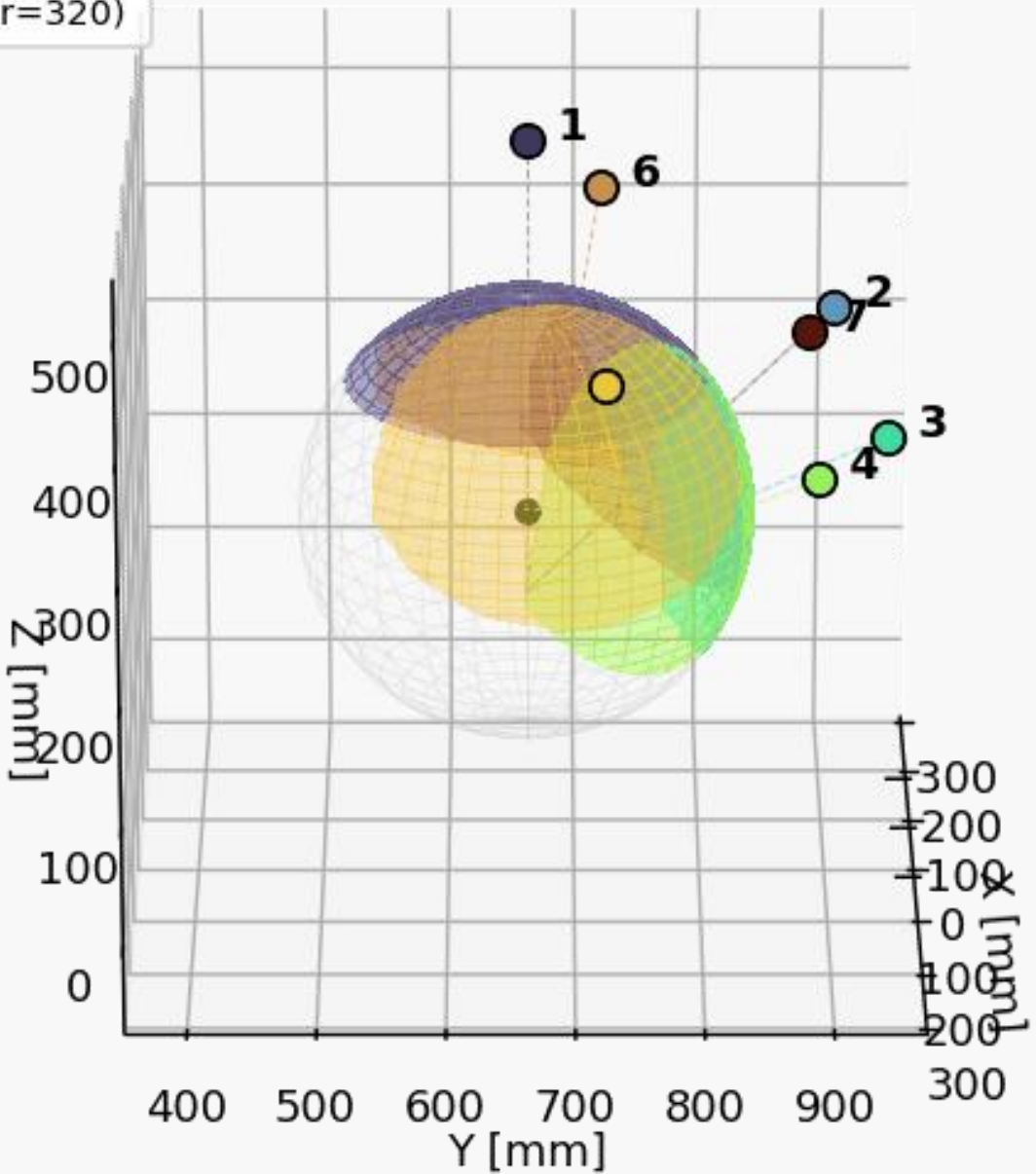
```
INSPECTION_POSITIONS = [  
    [r, 0, 90],    # top  
    [r, 0, 45],    # front1  
    [r, 0, 25],    # front2  
    [r, 45, 45],   # lat dx1  
    [r, 40, 80],   # lat dx2  
    [r, -40, 80],  # lat sx2  
    [r, -45, 45],  # lat sx1  
]
```

`im.point_and_shoot([r,  $\alpha$ ,  $\beta$ ])`

```
move_circle_spherical([r,  $\alpha$ ,  $\beta$ ])  
  
#image stabilization  
time.sleep(1)  
  
take_defects_global()  
  
defect_list.append()
```

capture (variable radius)

- 1: a=0, b=90 (r=380)
- 2: a=0, b=45 (r=340)
- 3: a=0, b=25 (r=314)
- 4: a=45, b=45 (r=320)
- 5: a=40, b=80 (r=346)
- 6: a=-40, b=80 (r=346)
- 7: a=-45, b=45 (r=320)





INSPECTION & MARKING

HELMET INSPECTION

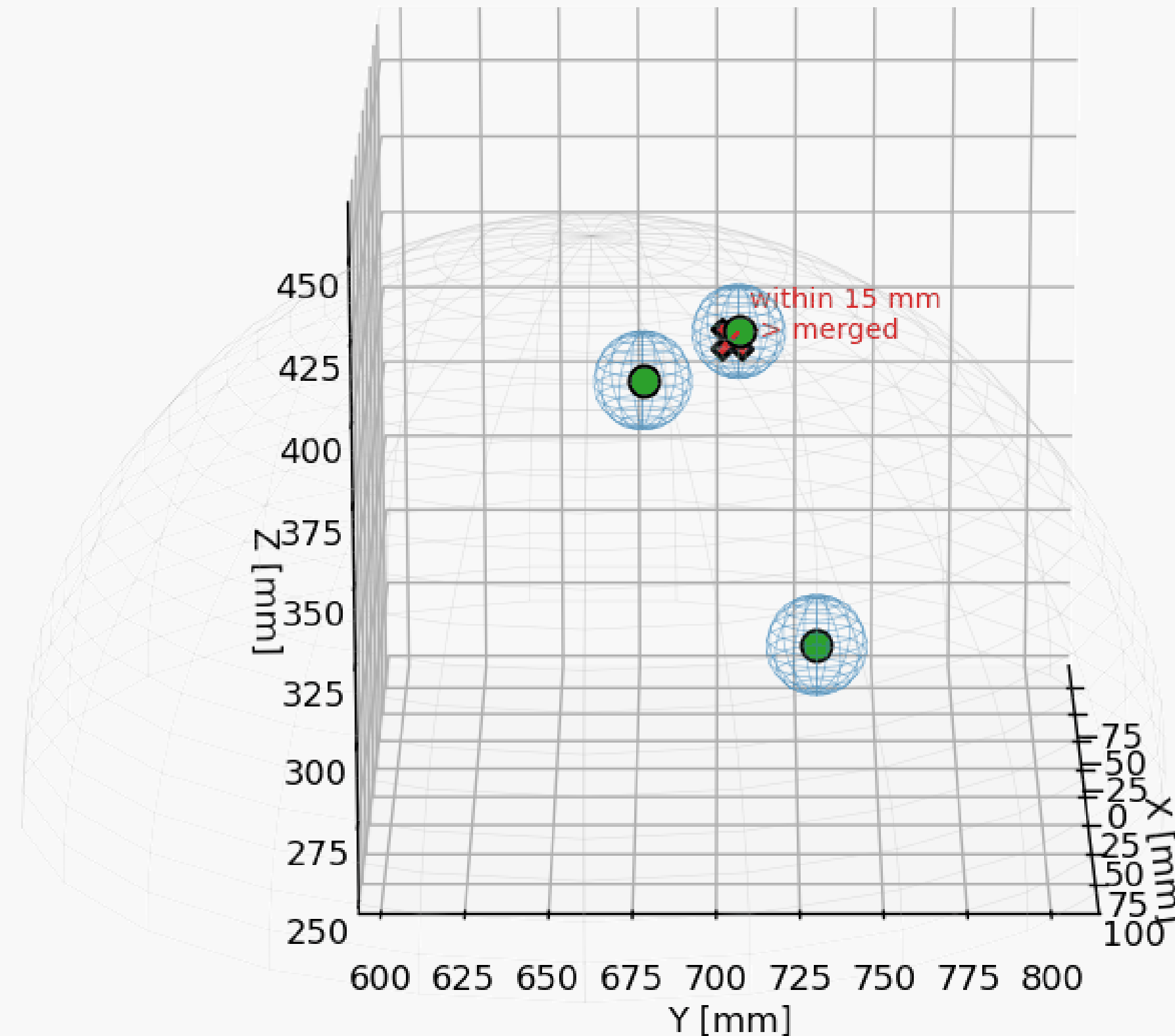
Filtering of defects identified multiple times

DUPLICATE FILTERING

By computing the distance between all defects, we can identify the ones that are too close (under a certain threshold) and keep only the one with the largest area.

```
im.duplicate_filter(defects)
```

```
for d in defect_list:  
    if dist < threshold:  
        keep largest area  
    unique.append(d)
```



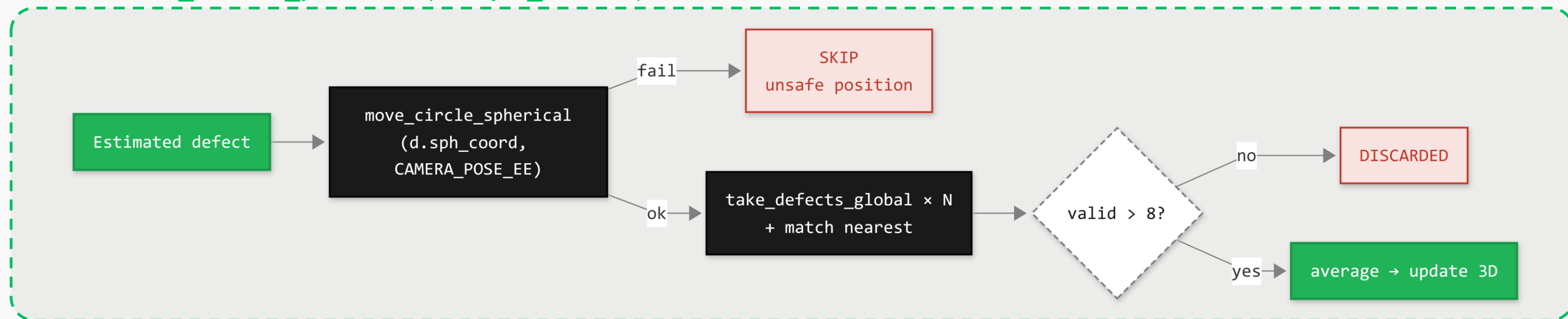


INSPECTION & MARKING

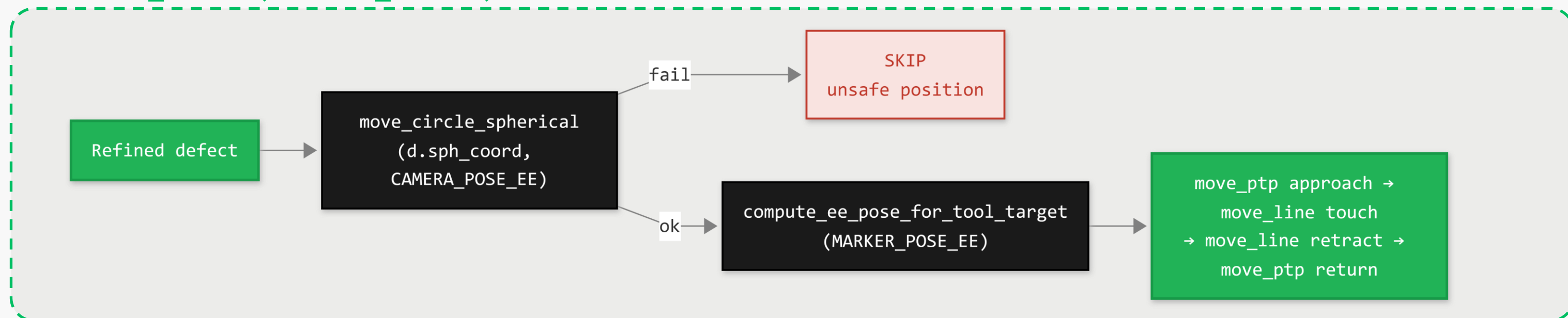
DEFECT REFINEMENT AND MARKING

for defect in unique_defects:

`im.refine_defect_position(unique_defect)`



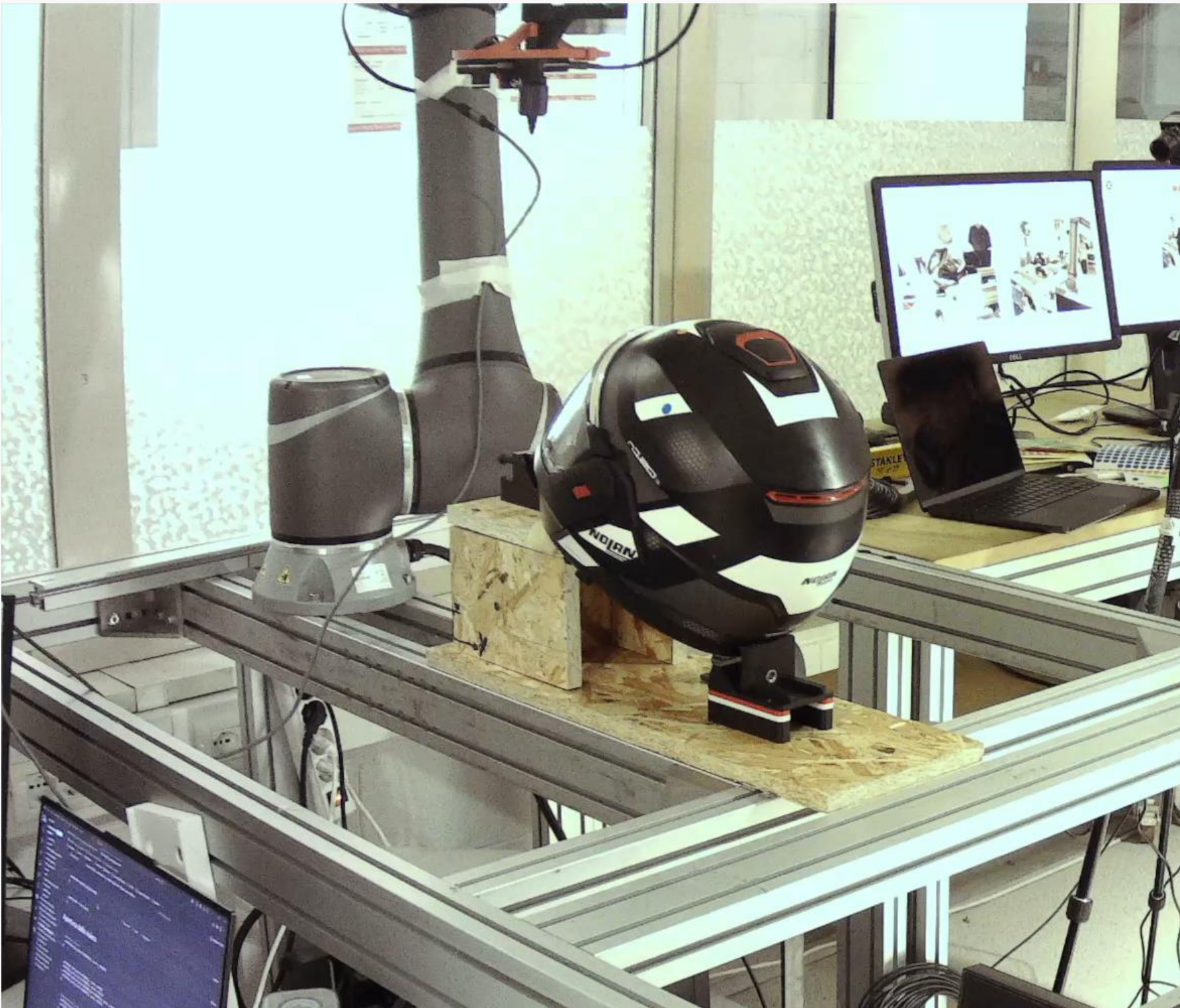
`im.mark_defect(refined_defect)`





INSPECTION & MARKING

DEFECT REFINEMENT AND MARKING



=== GLOBAL INSPECTION ===
Number of inspection positions: 7



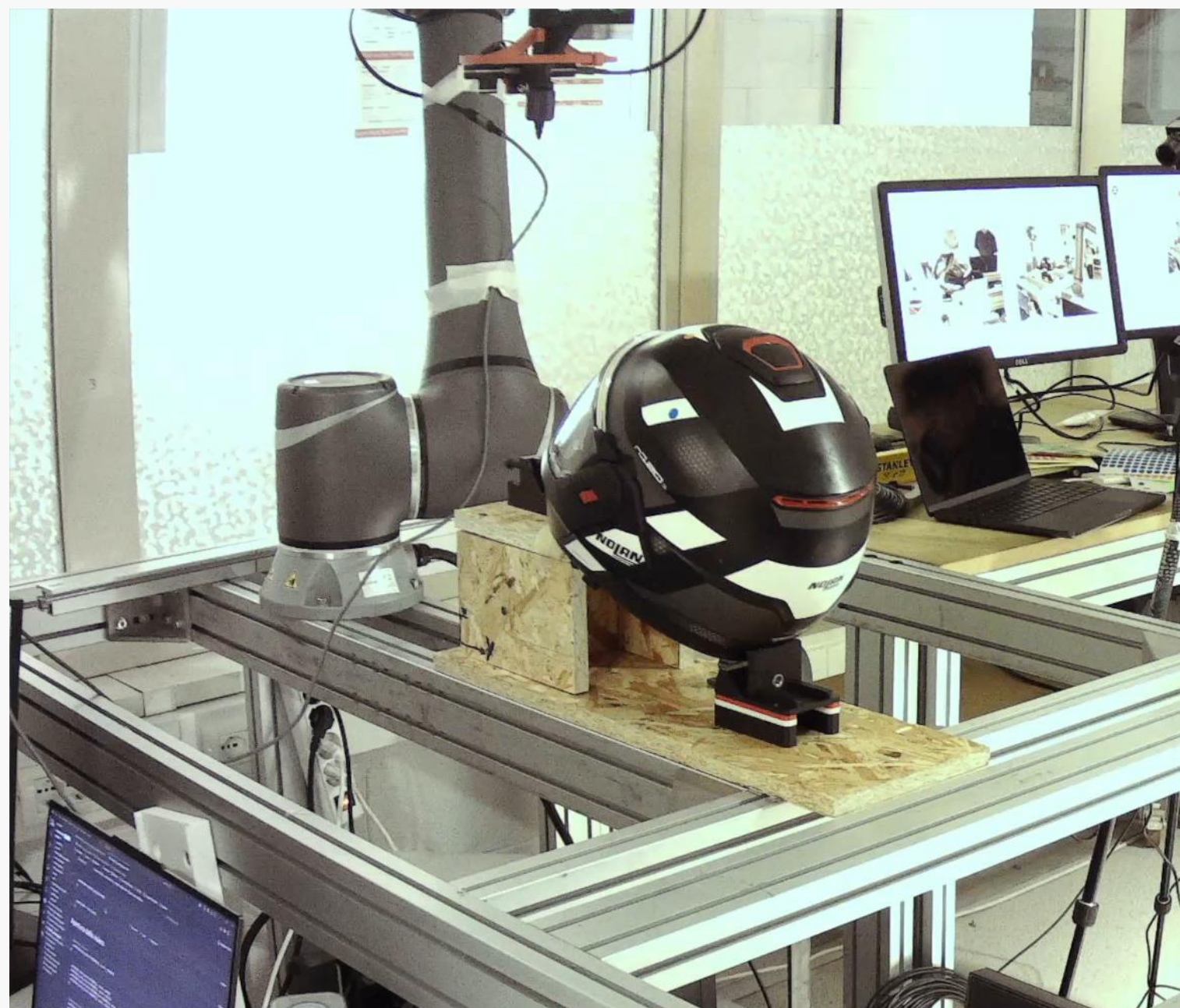


POLITECNICO
MILANO 1863

WRAP-UP

CONCLUSIONS

A complete and repeatable inspection cell



01 EASY AND FAST FIXTURING

Modular & tool-free, fits both helmet sizes

02 INTEGRATED END-EFFECTOR

Handling, vision, lighting & marking in one tool

03 VISION PIPELINE

Colored-defects detection located in 3D

04 AUTOMATIC

Detect, refine and mark automatically

